AI-Driven Personal Finance Management System – A Survey

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Abstract - Artificial intelligence is significantly reshaping financial management, enabling new levels of automation, personalization, and predictive insight for both individuals and organizations. As digital financial services expand, AI-driven tools are increasingly used for budgeting, investment management, risk assessment, and tracking financial goals. This paper explores the convergence of AI and financial technology by reviewing recent research, core functionalities, and real-world implementations. The analysis identifies major challenges like data privacy, algorithmic bias, integration complexity, and ethical considerations, which must be addressed for responsible adoption. While AI offers significant benefits in efficiency and user empowerment, the future of personal finance management will depend on advancing transparency, security, and inclusivity to ensure these systems are both effective and reliable.

Index Terms - Personal Finance Management, FinTech, Machine Learning, Predictive Analytics in Finance, Artificial Intelligence

I. INTRODUCTION

It is evident that finance has become an essential foundation of modern society, governing the handling, distribution, and application of monetary resources at both individual and institutional levels. The financial domain spans a broad spectrum of activities, including banking, investment, budgeting, lending, insurance, and economic policy. These activities support daily life functions such as salary payments, bill management, credit access, and long-term wealth planning, while also enabling corporations and governments to make decisions about resource distribution, infrastructure development and economic growth.

In everyday scenarios, individuals indulge in financial decision-making when budgeting expenses, managing savings, taking loans, or investing in assets. On a larger scale, businesses depend on financial operations for liquidity management, revenue forecasting, risk mitigation, and capital optimization. The macroeconomic role of finance is equally critical, as it facilitates the functioning of markets, supports innovation, and maintains economic stability. Thus, finance is not only a technical function but also a socio-economic infrastructure that supports national productivity and global interconnectivity.

With increasing digitalization, financial services have expanded rapidly through the growth of web-based financial applications. These platforms offer a wide range of services such as mobile banking, online trading, personal finance tracking, peer-to-peer lending, insurance management, robo-advisory, and digital wallets. Such applications have transformed traditional financial operations by providing seamless, real-time, and user-friendly interfaces that cater to both individual consumers and institutional clients. As financial transactions are data intensive, these systems must handle large volumes of structured and unstructured data, often under strict regulatory and performance constraints.

To meet these demands, Artificial Intelligence (AI) has emerged as a leader in the evolution of financial technology (FinTech). AI-driven systems can learn from past and present data to support activities such as fraud detection, false transactions, credit scoring and algorithmic trading. Machine learning (ML) techniques, for instance, allow applications to identify anomalies in transaction behavior, enabling proactive fraud prevention **[1]**. Deep learning (DL) models are employed to analyze and categorize expenses from textual data, while reinforcement learning (RL) methods are increasingly used to automate investment and budgeting strategies that adapt to uncertain market conditions.

The significance of AI in finance is further underscored by the fact that banking, financial services, and insurance are among the automated advisory services, as shown below in **Fig.1.** This widespread adoption highlights the central role of AI in transforming financial operations and driving innovation across the sector.

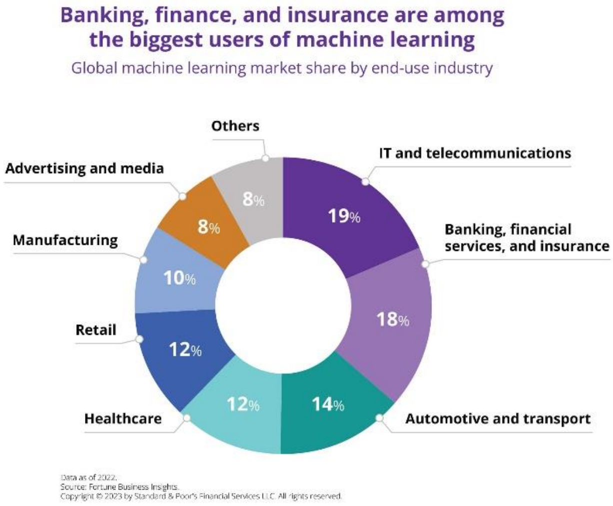

Fig.1 Global industry users of Machine Learning **[2]**

However, integrating AI into financial applications introduces several challenges. These include data privacy and security, the interpretability of AI models, compliance with financial regulations, and the need towards robust systems in unpredictable financial environments. Moreover, traditional ML algorithms assume static datasets that can be entirely loaded into memory, often fall short when faced with high volume, velocity, and variety of financial data in real-world systems. As a result, there is a growing need for scalable, explainable, and regulatory-compliant AI models tailored for financial applications.
The objective of this paper is twofold. First, to explore the convergence of AI and financial technology by highlighting key research developments, current methodologies, and practical use cases. Second, to analyze the emerging challenges and trends associated with AI-driven financial applications, including ethical, regulatory, and technical considerations

II. LITERATURE OF AI ON PERSONAL FINANCE

A. Understanding the Role of AI in Personal Finance

Over the past decade, AI has progressively emerged as a key driver in personal finance management. While early digital tools were limited to simple functions like manual expense tracking and basic budgeting, the integration of AI has enabled much more advanced capabilities. Today, AI is embedded in a variety of personal finance solutions, from automated investment platforms and digital advisors to smart budgeting apps and virtual assistants. The current landscape is characterized by the rapid expansion of fintech companies and digital banking services, which increasingly rely on AI to deliver more user-friendly, effective, and tailored financial experiences to users.

Several important factors have fueled the adoption of AI in personal finance. The explosion of digital transaction data, improvements in machine learning techniques, and growing consumer expectations for convenience and customization have all played a role. People now look for instant, tailored financial advice and seamless automation in managing their money. Meanwhile, financial institutions are leveraging AI to streamline operations, reduce costs, and enhance customer engagement. These trends are driving the industry toward smarter, more adaptive, and widely accessible financial tools.

B. Core Features and Functionalities Enabled by AI

The essential features of a modern personal finance app are illustrated in **Fig.2**, which highlights the core components that contribute to an effective and user-centric financial management experience. Among these, several key features are particularly significant for AI-driven personal finance applications and are discussed in detail below.

i. *Automated Budgeting and Expense Tracking*
AI-driven personal finance tools can classify expenses in real time, monitor spending habits, and generate real-time budget summaries, making it simpler for users to manage their daily finances. For instance, there are applications that streamline budgeting for minimalists by using machine learning to track and classify expenses with minimal user input [6]. Many leading budgeting apps now leverage AI for transaction tracking and automated expense categorization, providing users with accessible and efficient financial management solutions **[9].**

ii. *Personalized Financial Recommendations*
Modern AI algorithms evaluate user financial preferences to deliver tailored advice on saving, spending, and investing. Some decision support systems use predictive modelling to offer customized budgeting and investment suggestions based on user data, helping users make more informed financial choices **[3]**. AI-driven advisors can also provide real-time, personalized recommendations that adapt to changing financial circumstances, enhancing the relevance and impact of financial guidance **[5]**.

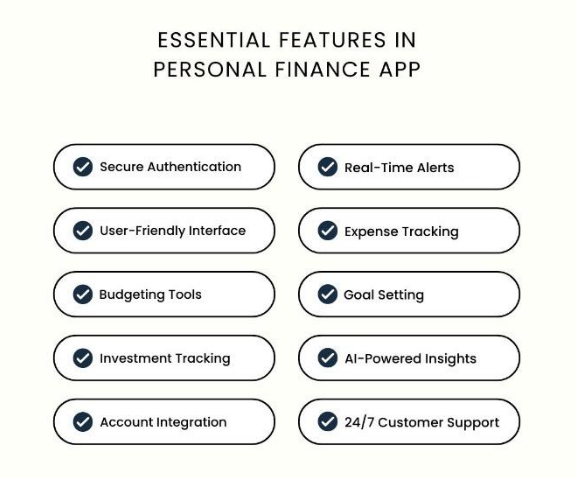

Fig.2 Must-have features in Personal Finance Apps

iii. *Investment Management and Robo-Advisory*
Robo-advisors powered by AI automate portfolio management, risk assessment, and asset allocation, making investment services more affordable and scalable for a wider audience. These platforms optimize portfolios and automate investment strategies for individual users, increasing accessibility to sophisticated financial planning **[4]**. Users of AI-driven investment tools often report higher satisfaction and improved returns compared to those relying on traditional advisory services **[11]**.

iv. *Risk Assessment and Fraud Detection*
AI models can identify irregular financial activities, assess credit risk, and identify potential fraud in real time, which significantly enhances financial security. Some financial planning tools use ML to evaluate financial risk and flag suspicious activities, improving both personal and corporate financial safety **[8]**. In enterprise financial systems, AI is used for anomaly detection and predictive risk analysis, enabling organizations to respond proactively to emerging threats **[7]**.

v. *Financial Goal Setting and Progress Monitoring*
AI-powered platforms assist users in setting realistic financial goals and tracking their progress through automated insights and reminders. Certain personal finance assistants leverage AI to help users define savings targets, monitor progress, and adjust strategies as needed **[10]**. Modern AI tools can also motivate users to achieve their financial objectives by providing ongoing feedback and adaptive recommendations **[12]**.

C. Empirical Studies and Real-World Implementations

Case Study 1: *Sly Spend – AI-Powered Personal Finance Management App*

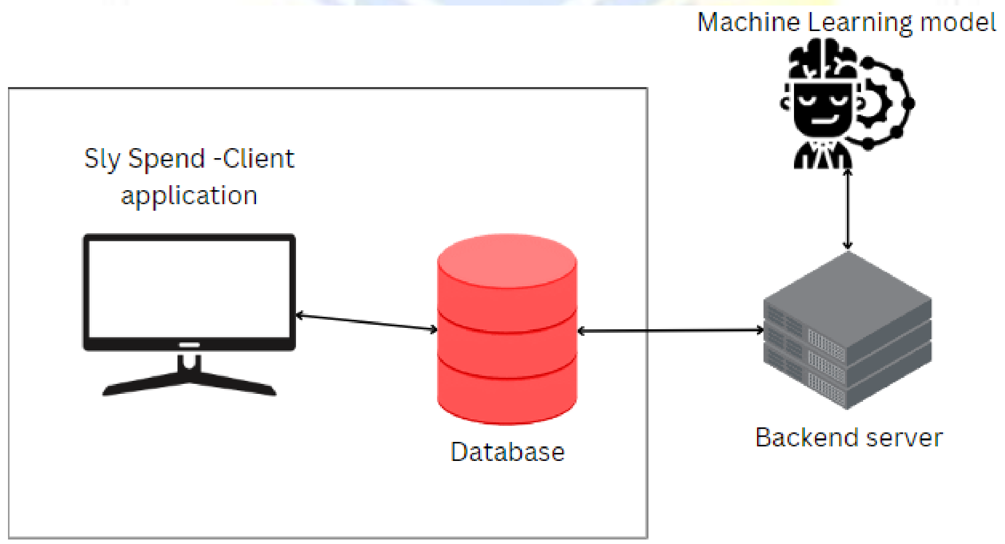

Fig.3 System Architecture of Sly Spend **[13]**

Sly Spend is a personal finance app that uses AI to help users monitor their income, expenses, and financial health. As shown in **Fig.3,** the system architecture consists of four main components: user interface, server infrastructure, database, and the machine learning (ML) model. The app connects users' financial accounts and sends transaction data to the backend server, where it is securely stored in the database. Its AI models—including linear regression, neural networks, and random forests—enable features such as personalized budgeting, goal setting, and financial risk assessment. Users can set savings or debt reduction goals, track their progress with interactive charts, and receive adaptive recommendations as the app learns from their behavior. Sly Spend also emphasizes user experience and security, offering an intuitive interface and robust data protection. Empirical results show that users benefit from improved spending control and financial awareness, with neural networks providing the most accurate predictions for future financial behavior **[13].**

Case Study 2: *Oracle EPM Cloud – AI-Driven Predictive Budgeting and Planning*
Oracle Enterprise Planning and Budgeting Cloud Service (EPBCS) integrates AI-driven automation into enterprise financial planning. The platform uses machine learning techniques such as time-series forecasting and regression analysis to generate more accurate and adaptive budget forecasts, drawing on both internal financial data and external economic indicators. Oracle EPM Cloud supports dynamic budgeting and rolling forecasts, enabling organizations to quickly respond to market changes. Its AI models detect anomalies in spending and highlight potential risks, while scenario planning tools help finance teams simulate different business conditions and optimize resource allocation. The system's modular design allows integration with various enterprise platforms and emphasizes data governance and model transparency. Organizations using Oracle EPM Cloud have reported improved budget accuracy and operational efficiency, provided they maintain high-quality data and robust governance practices **[7]**.

D. Comparative Analysis of Key Features in AI-Driven Personal Finance Systems

A comparative assessment of recent AI-driven personal finance systems is presented in Table 1. Five essential features are evaluated across four representative implementations to highlight both maturity and the gaps in current solutions.

Feature / **System**	Sly Spend **[13]**	Agarwal et al. **[10]**	MyFinanceAI **[14]**	Oracle EPBCS **[7]**
Automated Budget Planner	✔	✔	✔	✔
Expenditure Logging & Categorization	✔	✔	✔	✘
Bill Split & Joint Budgeting	✘	✘	✘	✘
Advisor Chatbot	✘	✔	✘	✘
Goal-Based Savings Tracker	✔	✘	✔	✘

Table 1. Comparative Feature Matrix of AI-Driven Personal Finance Systems

Despite broad support for budgeting and expenditure tracking, advanced features such as bill splitting, advisor chatbots, and goal-based savings remain limited or absent in most implementations. This comparative analysis reveals that current AI-driven personal finance systems are yet to fully address collaborative financial management and interactive user guidance. These persistent gaps highlight the need for future research to focus on developing more comprehensive, user-centric solutions that can support shared budgeting scenarios and provide proactive, personalized financial advice. The subsequent section addresses the principal challenges facing the adoption and evolution of AI-driven personal finance management.

III. CHALLENGES

AI-powered personal finance tools are reshaping how people handle their money, yet their growing use presents unique hurdles that require thoughtful solutions for ethical and efficient application.

A primary concern is safeguarding sensitive financial details, as these systems manage critical user information, exposing them to potential cyber risks. Strengthening defenses with sophisticated encryption, decentralized learning approaches, and adherence to privacy standards is crucial to protect data and build user reliability.

Bias and fairness in AI models present another significant concern. If trained on historical or unrepresentative data, these systems can worsen the existing biases, leading to unfair or discriminatory financial advice. Ongoing research is focused on developing fairness-aware algorithms and using diverse datasets to overcome such risks.

The lack of human sensitivity poses a significant challenge, as AI excels at data processing but struggles to mimic the empathy and deep understanding human advisors provide during stressful financial moments. Developments in language processing and emotional analysis are being pursued to improve AI's human-like interaction.
Over-reliance and financial literacy erosion is an emerging challenge. As users increasingly depend on AI for financial decisions, there is a risk that their own financial knowledge and decision-making skills may decline. Researchers are addressing this by designing AI tools that also educate users, helping them understand financial concepts and the reasoning behind recommendations.

Integration and data silos complicate the effectiveness of AI-driven advisors. These systems often need to aggregate data from multiple financial institutions, but inconsistent standards and siloed data can hinder seamless integration. Open banking initiatives and secure API development are key research areas to promote interoperability.

Scope and complexity limitations also restrict the capabilities of AI advisors. While AI excels at well-defined tasks, it struggles with complex, long-term financial planning that requires human judgment and expertise. Hybrid intelligence models and advancements in AI reasoning are being explored to address these multifaceted challenges.

Finally, ethical considerations beyond bias are increasingly important. Issues such as algorithmic nudging, potential market instability from herd behavior, and the digital divide that may exclude certain users from accessing AI-driven financial services highlight the need for comprehensive ethical frameworks and societal impact assessments.

Overcoming these challenges is key to ensuring that AI-driven personal finance advisors remain secure, equitable, and widely accessible. Sustained attention to these issues will enhance user confidence and the overall reliability of digital financial solutions.

IV. FUTURE SCOPE

Looking ahead, the next generation of AI-powered personal finance systems is likely to focus on enhancing privacy, transparency, and user control. Future platforms are expected to adopt advanced privacy-preserving techniques, allowing users to benefit from personalized financial insights without compromising sensitive data. The integration of open banking frameworks and secure data-sharing protocols will enable users to seamlessly manage multiple accounts and financial products within a single, unified interface. As these systems become more interconnected, real-time data analysis and adaptive budgeting features will help users respond proactively to changes in their financial situation.

Another important direction is the development of more transparent and explainable AI models that can clearly communicate the reasoning behind financial recommendations. This will not only build user trust but also support compliance with evolving regulatory standards. Additionally, combining AI-driven automation with human expertise—especially for complex or long-term financial planning—will help address the limitations of fully automated systems. By focusing on these areas, future personal finance solutions can become more secure, user-friendly, and adaptable, empowering a broader range of individuals to achieve their financial goals with confidence.

V. CONCLUSION

The rapid evolution of artificial intelligence is redefining the landscape of personal finance, offering individuals unprecedented tools for managing, planning, and optimizing their financial lives. As AI-driven systems become more sophisticated, they are making financial management more accessible, efficient, and tailored to individual needs. However, realizing the full potential of these technologies requires careful attention to challenges such as data privacy, ethical considerations, and the need for transparency. By embracing innovation while prioritizing user trust and responsible design, the next generation of AI-powered personal finance solutions can support users in achieving greater financial security, literacy, and empowerment in a dynamic digital era.

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